WHAT THE WELD'S NON-ASSOCIATIVITY IS — AND IS *NOT*: A TRIANGULATION ACROSS FOUR OBSTRUCTION THEORIES, AND THE IDEMPOTENT NORMALIZER THAT RECEIVES IT

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Abstract. The worker/harness *weld* — the Zappa–Szép (orchestration) composition of directed containers, the program's central tool for *when* compositions exist — is non-associative in general. We ask which categorical invariant measures that non-associativity, and answer in two movements. First, by *negative triangulation*: four *named* candidate maps are refuted, each in a structurally distinct way. The renormalization idempotent $k = n\,;m$ of Di Lavore–Román–Székely is not a 2-cocycle (no receptacle: the cocycle condition *is* associativity), not a Karoubi/Cauchy splitting obstruction (vacuous receptacle: k already splits), and not a lax distributive law of Schoen/Tholen (wrong locus: laxness is a fixed-direction inequality on a coherence cell, whereas the associator defect compares two *order-incomparable* normalized distributions). Second, and positively: the surviving invariant is received *exactly* by an idempotent *normalizer* K — a hom-operator obeying a left- but not right-absorption law. Writing $f \# g := K(f \cdot g)$, associativity of $\#$ is equivalent to K being two-sided absorbing, so the handedness obstruction is genuine and non-vacuous; and it is provably *operator-level*, never *element-level*, which refutes the fourth named map — the near-semiring's one-sided-absorbing *zero*. The normalized-kernel magmoid **Norm** is an instance, and a machine-checked finite model certifies the construction axiom-free. The originality of the surviving invariant is reported honestly, as the current state of a literature sweep, not as a theorem.

1. Setup: the weld and its non-associativity

Throughout, **Poly** is the category of polynomial functors on **Set** with the composition monoidal product $\triangleleft$; a comonoid for $\triangleleft$ is a *polynomial comonad*, equivalently a small category. For a set S we write $\Delta S = S\mathsf{y}^S$ (notation y for the representable) for the *store* comonad on S, the directed container of the full state space on S. All composites are written in diagrammatic order: $f\,;g$ (equivalently $f \cdot g$) means "f then g", and vertical composition of natural transformations is likewise diagrammatic.

Definition 1.1 (The weld)**.** Given two directed containers presented by store comonads ΔS (the *worker* state) and ΔT (the *harness* state), their *weld* is the orchestration composite

$$B \;=\; \Delta S \triangleleft \Delta T,$$

the Zappa–Szép product that interleaves worker and harness transitions. The weld is the program's canonical answer to "when does a composite of two stateful directed containers exist?": the Zappa–Szép datum is exactly the mutual-action data that makes the interleaving well defined.

The weld is *not* associative in general. This is not a defect of our encoding; it is intrinsic. The sharpest extant witness is probabilistic. Di Lavore, Román and Székely ("Normalized Probabilistic Semantics is Not Associative", arXiv:2510.01131) show that *normalization* of substochastic kernels is a *sesquilaw*, not a distributive law, and that the induced Kleisli-style composition of normalized kernels is a genuine *magmoid* — a category minus associativity.

Definition 1.2 (Sesquilaw and the renormalization idempotent, after arXiv:2510.01131)**.** A *sesquilaw* (S, T, m, n) consists of a distributive law $m\colon ST \Rightarrow TS$ together with an *almost-distributive* law

Date: 19–20 October 2026.

[0]MacBeth (scotmacbeth) · 2026-10-19/20 · work-in-progress commit: `ac07a05`

$n\colon TS \Rightarrow ST$ satisfying all four distributive-law axioms *except* multiplicativity, which holds only up to the natural transformation

$$k \;=\; n\,;m \;:\; TS \Rightarrow TS \qquad \text{(the "renormalization" of Lemma 3.5).}$$

For normalization one takes $T = D$ (distribution) and $S = M$ (maybe), so $TS = DM$ and $ST = MD$; then $n = (-)^{\circ}\colon DM \Rightarrow MD$ is normalization and $m = (-)^{\bullet}\colon MD \Rightarrow DM$ is the subdistribution inclusion. The induced magmoid **Norm** of normalized kernels $X \to MDY$ is a category minus associativity; Proposition 4.9 of loc. cit. isolates the triples that do associate.

Remark 1.3. We treat $k = n\,;m$ as a faithful, finite-dimensional stand-in for the weld's associativity defect. The identification is justified computationally below (§9, reproducing arXiv:2510.01131's Prop. 4.9 purity cases exactly) and at the level of principle by the fact that both objects are instances of "an idempotent measuring departure from a distributive law." The question of this note is: *which categorical invariant is k (equivalently, the weld's non-associativity)?*

2. The method: name the map, then decide

We adopt one discipline, learned the hard way across a month of cross-domain bridge-hopes that died on inspection.

Remark 2.1 (Naming the map buys decidability, not survival). A candidate obstruction home ranked highly because its *vocabulary* matches ("cocycle", "splitting", "Yang–Baxter") has earned nothing until a *structure map* is exhibited between the *specific* objects at hand — not the words. "This sounds like it should connect" is a prior of roughly zero. The payoff of naming a candidate map is not that it survives; it is that the question becomes *decidable*. Our record on resonance-ranked leads is $0/n$; our record on *named*-map leads is: all decided, crisply, one way or the other.

The note has two movements. In the first (§§3–6) we record the one named map that lands (the associative corner) and refute three of the four candidate obstruction homes. In the second (§§7–11) we exhibit the structure that *does* receive the invariant, and show that building it refutes the fourth candidate as a by-product. Each verdict is one of three epistemic kinds, kept strictly separate:

- **proved** — a registered result of the program (the four verdicts and the positive construction; several carry Lean-verified leaves);
- **evidence** — a literature-sweep conclusion, *not* a theorem (the handedness-originality claim, §11);
- **computed** — a corroborating but non-formalized numerical fact (so flagged in place).

Movement I. Four homes it is not

3. The associative corner: the weld is a regular bicomodule (EXISTS)

Before the refutations, we record the one named map that *does* land — so the reader sees the method is not merely destructive, and so the associative regime is kept firmly separate from the defect we chase.

Theorem 3.1 (Spivak–Garner–Fairbanks, "Functorial Aggregation", arXiv:2111.10968, Prop. 3.9(5) + Thm. 4.28)**.** *For polynomial comonoids (equivalently small categories) c and d, there is an equivalence*

$$\mathrm{Comod}(\mathbf{Poly})(c,d) \;\simeq\; \mathrm{PRA}(c\text{-}\mathbf{Set},\, d\text{-}\mathbf{Set})$$

between (c,d)-bicomodules in **Poly** *and parametric right adjoint functors c-**Set** $\to$ d-**Set**.*

Proposition 3.2 (weld is a regular bicomodule; registry `weld-is-regular-bicomodule`, **proved**)**.** *The weld $B = \Delta S \triangleleft \Delta T$ is the* regular $(\Delta T, \Delta S)$*-bicomodule. Under Theorem 3.1 it represents the prafunctor*

$$\bar{F}_B(V) \;=\; (T \times V)^S \;=\; T^S \cdot V^S,$$

whose slice (aggregation) factorization keeps the power $(-)^S$ as connected-limit content and aggregates over the branch-map set T^S by a coproduct Σ.

This result is peer-reviewed within the program (Rick's referee pass, rounds 1 and 2) and carries a Lean-verified leaf checking the five regular-bicomodule coaction axioms.

Remark 3.3 (The bicomodule is associative; keep it separate from the defect)*.* Crucially, the bicomodule/PRA picture of Proposition 3.2 lives in the *associative* regime. The weld, *once its Zappa–Szép data make it well defined*, is a genuine comodule, and the cohomological class $[\omega] \in H^2$ attached to it (elsewhere in the program) is a *splitting* obstruction of an *already-associative* weld, not a measure of non-associativity. The non-associativity we chase here is the Kleisli-side sesquilaw defect — the *departure* from that associative regime. This separation is the hinge of the whole note.

4. Refutation I: k is not a 2-cocycle (no receptacle)

Claim 4.1 (registry: refuted; cf. `h2-weld-class-is-splitting-not-associativity`)*.* The renormalization idempotent $k = n\,;m$ is *not* a 2-cocycle. The refutation is not that the putative class vanishes; it is that the *receptacle does not exist.*

Why the receptacle is empty. A 2-cocycle condition, in every incarnation (group cohomology, the Sweedler/Hopf cocycle, the obstruction cocycle of an extension), *is* an associativity equation: the cocycle identity is precisely the statement that two bracketings of a triple agree. Thus a 2-cocycle presupposes an associative multiplication whose coherence it then measures. A genuinely non-associative magmad has no associative multiplication to begin with, hence no cocycle at all — there is nothing for a class to be a class *of*. One cannot carry the non-associativity as a cohomology class because the apparatus that would carry it is *defined by* associativity. (Confirmed independently by Román's magmads: associativity-failure is a *magma* phenomenon, not a cocycle phenomenon.) □

This is a *well-posed negative of the first kind*: the map was named, checked, and the home turned out to be empty. Its locus is the *cochain* (single-morphism) level.

5. Refutation II: k is not a Karoubi splitting obstruction (vacuous receptacle)

Here the receptacle *exists*, and that makes the negative subtler and, we think, more instructive.

Lemma 5.1 (k lives in an associative hom-set and is idempotent)**.** *$n\colon DM \Rightarrow MD$ and $m\colon MD \Rightarrow DM$ are morphisms in the functor category $[\mathbf{Set}, \mathbf{Set}]$, whose composition (vertical composition of natural transformations) is strictly associative. Hence $k = n\,;m \in \mathrm{End}_{[\mathbf{Set},\mathbf{Set}]}(DM)$, and "is k idempotent?" and "does k split?" are both well posed. Moreover k is idempotent: from the* section identity *$m\,;n = \mathrm{id}_{MD}$ (a normalized distribution, included as a subdistribution, already has unit mass, so renormalizing is the identity),*

$$k\,;k \;=\; (n\,;m)\,;(n\,;m) \;=\; n\,;(m\,;n)\,;m \;=\; n\,;\mathrm{id}_{MD}\,;m \;=\; n\,;m \;=\; k.$$

Theorem 5.2 (sesquilaw k is a split idempotent; registry `sesquilaw-magmoid-karoubi`, **proved**)**.** *$k = n\,;m$ is a* split *idempotent, split by*

$$DM \xrightarrow{\;n\;} MD \xrightarrow{\;m\;} DM, \qquad m\,;n = \mathrm{id}_{MD}, \qquad n\,;m = k,$$

with splitting object the magmad carrier MD itself. Consequently:

(1) *the Karoubi/Cauchy (idempotent) completion is a* no-op *on k — k already lies in the image of* $[\mathbf{Set}, \mathbf{Set}] \hookrightarrow \mathrm{Kar}([\mathbf{Set}, \mathbf{Set}])$*, so there is no "missing splitting";*
(2) *there is* no splitting obstruction *to carry the non-associativity: the feared failure (that k does not split) is empty.*

Proof. A split idempotent is exactly a factorization $e = s\,;r$ with $r\,;s = \mathrm{id}$ through the image; Lemma 5.1 exhibits $r = n$, $s = m$, with $m\,;n = \mathrm{id}_{MD}$ and $n\,;m = k$. The splitting object MD is already present in $[\mathbf{Set}, \mathbf{Set}]$, so idempotent completion adds nothing for k. Both consequences follow. □

What makes this decisive is the meta-lemma it instantiates.

Lemma 5.3 (idempotent completion preserves associativity)**.** *For any category $\mathcal{D}$, the Karoubi/Cauchy completion* $\mathrm{Kar}(\mathcal{D})$ *is a* category*: its composition is associative (Borceux I, §6.5; nLab, "idempotent completion").*

Corollary 5.4 (splitting can never measure non-associativity)**.** *Splitting idempotents in an associative $\mathcal{D}$ yields only an* associative *category. Hence no "split some idempotents in an associative core" construction can* produce *a non-associative composition; dually, the failure of a composition to be associative can never be* identified with *the failure of an idempotent to split, because the latter is repaired by a construction (Lemma 5.3) that cannot repair the former.*

Corollary 5.4 is why this candidate was doomed *before* any computation: even had k failed to split, completing to repair it would land in an associative category, so the repaired object could not witness the magmoid's genuine non-associativity. This is a *well-posed negative of the second kind*: the home exists but carries no obstruction content. Its locus is the *single idempotent morphism.*

The Karoubi vacuity is generic, not weld-specific. An independent source, from a problem with no connection to the weld, confirms that this vacuity is a general feature of the weak-distributive-law family rather than an artifact of our particular k. Böhm–Lack–Street ("Idempotent Splittings, Colimit Completion, and Weak Aspects of the Theory of Monads", arXiv:1102.4931) show that weak monad notions — e.g. *demimonads*, which drop the unit law but keep associativity $\beta \cdot T\beta = \beta \cdot \mu$ — reduce systematically to *strict* structure inside the Cauchy-completed $Q_*\mathcal{K}$: an independent, pre-weld proof that Cauchy/Karoubi completion *absorbs* one-sided weakenings of the monad laws rather than *detecting* them, exactly Corollary 5.4 reached from a wholly different problem. This upgrades the verdict from "true for our k" to "structurally true for the family". (A second, probabilistic echo — that idempotent Markov kernels generically split — is recorded in the program's reading log at summary grade, flagged for a deep read before it is cited.)

6. Refutation III: k is not a lax distributive law (wrong locus)

The one technical (non-vocabulary) lead for a prior home of one-sidedness is *lax distributive laws by inequality* — Schoen, "Lax Liftings and Lax Distributive Laws" (arXiv:2307.16552, ACT 2022), and the companion (T, V)-program of Tholen, "Lax Distributive Laws for Topology I/II" (arXiv:1603.06251, 1605.04489). We decide it, and it is the subtlest negative: the home exists, is inhabited, and is genuinely one-sided — but one-sided *about the wrong thing.*

Definition 6.1 (lax distributive law, after Schoen)**.** Let P be a monad on $\mathbf{Set}$ and F an *endofunctor* (no algebraic structure). A *lax distributive law* is a transformation $\lambda_X \colon FP(X) \to PF(X)$ subject to three *one-directional inequalities* in the inclusion order $\subseteq$ of relations (here $\mathbf{Set}$ is order-enriched to $\mathbf{Rel}$, with $\mathbf{Rel}(X, Y) = P(X \times Y)$ a poset under $\subseteq$): lax naturality, and lax compatibility with the unit η and multiplication μ of P — the last two pointing in *opposite* directions by design. The laxified cells are the coherences of the *one* monad P; F is a bare functor and contributes none.

The one-sidedness of a lax distributive law is a *fixed-direction order inequality on a coherence cell*. The weld's one-sidedness is of a different grammar, and the mismatch is a theorem, not an impression.

Claim 6.2 (registry: refuted; `handedness-not-lax-DL` **proved**). The weld's handedness is not an instance of, and is not faithfully receivable by, a Schoen/Tholen lax distributive law.

We give two independent reasons. The first is bookkeeping; the second is the conceptual kill.

Variance. Schoen's template needs a triple (P monad, F bare functor, $\lambda\colon FP \to PF$); the weld supplies (D, M, m, n, k). Only two variance assignments type-check. With $P = M,\ F = D,\ \lambda = n\colon DM \to MD$, the map n does carry the defect — but its failing coherence is D-multiplicativity, a coherence with the multiplication of $T = D$, which in Schoen's template (where $F = D$ is a *bare* functor) is *invisible*; the three laxified cells are all about $P = M$. With $P = D,\ F = M$, the only map of the right variance is the *strict* law m, which carries no defect; the defect-carrying n has the wrong variance. In both assignments the handedness defect is not among the laxifiable cells. No structure map exists; the same analysis applies verbatim to Tholen (same coherence-cell laxness, quantale-enriched).

The discriminator. The conceptual kill is one sentence and is independent of the variance bookkeeping.

Lemma 6.3 (elementary). *If $\mu \le \nu$ pointwise for probability measures μ, ν on a set Y (total mass $\sum \mu = \sum \nu = 1$), then $\mu = \nu$.*

Proof. $\nu - \mu \ge 0$ pointwise and $\sum(\nu - \mu) = 1 - 1 = 0$; a nonnegative function with zero total is zero. □

Proposition 6.4 (handedness is not a laxness). *The associator defect of* **Norm** *is not expressible as a fixed-direction lax inequality in the canonical (pointwise sub-probability) order-enrichment — the order under which the Schoen/Tholen framework applies.*

Proof. For a triple (f, g, h) whose composite succeeds, both bracketings $(f \# g) \# h$ and $f \# (g \# h)$ lie in **Norm**'s hom-set, i.e. are *normalized* distributions of total mass 1 (the outer normalization $(-)^\circ$; cf. Definition 1.2). A Schoen/Tholen laxness would require a *single fixed direction* — $(f \# g) \# h \le f \# (g \# h)$ for all triples, or the reverse for all triples. But whenever the two bracketings differ, Lemma 6.3 forbids them from being $\le$-comparable; and non-associativity is precisely the regime where they differ. So neither fixed direction can hold: the defect lives strictly between $\le$-incomparable elements and is not a laxness. □

The honest near-miss, dispatched. There *is* a genuine lax cell in the DM order, namely $\mathrm{id}_{DM} \le k$ pointwise (normalization raises mass: $k(\mu)(y) = \mu(y)/|\mu| \ge \mu(y)$ since $|\mu| \le 1$). But that cell is a statement about D's renormalization, not one of Schoen's $P = M$-coherences, and it is not *what the handedness is*. Handedness is the statement "k absorbs in $\#$'s left slot but not the right" — a fact about $\#$ as a binary operation — and Proposition 6.4 shows that fact cannot be any fixed-direction $\le$. A lax cell existing *somewhere* in the ambient order does not make the *associator defect* a laxness.

This is a *well-posed negative of the third kind*: the home exists and is one-sided, but its one-sidedness is a fixed-direction $\le$ on a *coherence cell* (one monad), whereas handedness is an asymmetry of the *argument slots* of the induced binary operation (two monads), with $\le$-incomparable outputs. The locus is the *coherence 2-cell*.

Remark 6.5 (honest scope of the Tholen half). Schoen arXiv:2307.16552 is deep-read; Tholen arXiv:1603.06251/1605.04489 is not directly read, and the no-match for Tholen rests on the *family-level* fact that its laxness is quantale-order laxness of coherence cells — the same locus Proposition 6.4 rules out — not on any fine Tholen detail. Were a future read to surface a Tholen

construction laxifying an *argument slot* of a composition rather than a coherence cell, this verdict should be revisited; we judge it unlikely but mark it as the one unread dependency.

Three named maps are now decided, of three distinct kinds — no receptacle (cocycle), vacuous receptacle (Karoubi), wrong-locus receptacle (lax-DL) — and each probes a level (cochain, idempotent morphism, coherence cell) at which, we will see, the invariant is simply not visible.

Movement II. The home it is

7. The positive home: an idempotent normalizer

If the defect is neither a cocycle, nor a splitting obstruction, nor a laxness, what *is* it? The honest answer is concrete, non-cohomological, and lives one level up from every map so far — at the binary operation itself. We isolate the minimal abstract structure that carries it. Two terms below, "normalizer" and "one-sided absorbing idempotent", are this note's own coinages, not standard vocabulary; we define them from scratch.

Fix a category $(\mathcal{A}, \cdot, 1)$ with strictly associative composition — think of the honest, associative ambient (substochastic kernels under matrix product; or the Kleisli category $\mathrm{Kl}(DM)$). Diagrammatic order: for $f \in \mathcal{A}(X,Y)$, $g \in \mathcal{A}(Y,Z)$, $f \cdot g \in \mathcal{A}(X,Z)$ is "f then g".

Definition 7.1 (normalizer, normalized magmoid)**.** A *normalizer* on $\mathcal{A}$ is a family of functions $K = K_{X,Y} \colon \mathcal{A}(X,Y) \to \mathcal{A}(X,Y)$ such that

- (K1) K is idempotent: $K(K(a)) = K(a)$;
- (U) K fixes identities: $K(1_X) = 1_X$;
- (LA) *left-absorption*: $K\big(K(x) \cdot c\big) = K(x \cdot c)$ for all composable x, c.

K is *two-sided* if additionally, for every $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} W$ with f, g, h in the image of K,

- ($\mathsf{RA}_{\#}$) *right-absorption for* $\#$: $K\big(f \cdot K(g \cdot h)\big) = K(f \cdot g \cdot h)$.

The *normalized magmoid* N_K has $\mathrm{Ob}\, N_K = \mathrm{Ob}\, \mathcal{A}$, $N_K(X,Y) := \mathrm{Im}\, K_{X,Y} = \{a : K(a) = a\}$, composition $f \# g := K(f \cdot g)$, and units $1_X^{\#} := 1_X$. A structure *with a one-sided absorbing idempotent* is $(\mathcal{A}, \cdot, K)$ with K a normalizer that is *not* two-sided.

Remark 7.2 (N_K is a magmoid, and associativity is not assumed). $f \# g = K(f \cdot g) \in \mathrm{Im}\, K$ by (K1), and (U) gives units: for $f \in N_K$ (so $K(f) = f$), $f \# 1_X = K(f \cdot 1) = K(f) = f$ and $1_X \# g = K(g) = g$. Associativity is *not* assumed — that is the whole point.

Remark 7.3 (the laws are not vacuous). In (LA) the argument x ranges over *all* of $\mathcal{A}(X,Y)$, in particular over products $x = f \cdot g$ of N_K-morphisms, which are generally *not* in $\mathrm{Im}\, K$ (composing two normalized kernels de-normalizes). Where $K(x) \neq x$, (LA) is a genuine constraint. Likewise ($\mathsf{RA}_{\#}$) concerns the *un-normalized* product $y = g \cdot h$; it is *not* the triviality $K(f \cdot K(y)) = K(f \cdot y)$ for $y \in \mathrm{Im}\, K$.

8. The characterization: associativity is two-sided absorption

Two theorems pin down N_K completely. Both use only $\cdot$ and K; neither invokes any 2-categorical machinery.

Theorem 8.1 (the left bracketing is clean)**.** *Let K be a normalizer. Then for all composable $f, g, h \in N_K$,*

$$(f \# g) \# h \;=\; K(f \cdot g \cdot h).$$

Proof. Expand and apply (LA) with $x := f \cdot g$:

$$(f \# g) \# h = K\big((f \# g) \cdot h\big) = K\big(K(f \cdot g) \cdot h\big) \overset{\text{(LA)}}{=} K\big((f \cdot g) \cdot h\big) = K(f \cdot g \cdot h),$$

the last step by associativity of $\cdot$ in $\mathcal{A}$. The left bracketing normalizes once, at the very end. □

The proof uses only (LA) and associativity of $\cdot$ — not ($\mathsf{RA}_\#$), not even the unit axiom. The right bracketing does not simplify this way, and the gap between them is the whole content.

Theorem 8.2 (associativity $\iff$ two-sided absorption)**.** *Let K be a normalizer. Then N_K is associative* if and only if *K is two-sided, i.e. ($\mathsf{RA}_\#$) holds.*

Proof. ($\Leftarrow$) Suppose ($\mathsf{RA}_\#$). For $f, g, h \in N_K$,

$$f \# (g \# h) = K\big(f \cdot (g \# h)\big) = K\big(f \cdot K(g \cdot h)\big) \overset{(\mathsf{RA}_\#)}{=} K(f \cdot g \cdot h) = (f \# g) \# h,$$

the last equality by Theorem 8.1. ($\Rightarrow$) Suppose $\#$ is associative. Theorem 8.1 gives $(f \# g) \# h = K(f \cdot g \cdot h)$, while $f \# (g \# h) = K(f \cdot K(g \cdot h))$ by definition; associativity equates the two, which is precisely ($\mathsf{RA}_\#$). □

Corollary 8.3 (handedness is a genuine, non-vacuous obstruction)**.** *If K satisfies (LA) but not ($\mathsf{RA}_\#$), then N_K is a* non-associative *magmoid, and its associator defect $(f\#g)\#h$ versus $f\#(g\#h)$ equals exactly the defect $K(f \cdot g \cdot h)$ versus $K(f \cdot K(g \cdot h))$ of right-absorption — nothing more and nothing less. Handedness = (LA) $\wedge \neg$($\mathsf{RA}_\#$).*

This is the sharp contrast with Refutation II. The Karoubi receptacle was *vacuous*: k already splits, so idempotent completion measures nothing (§5). Here the receptacle is *inhabited*: Corollary 8.3 identifies a concrete, nonempty set of triples — those with $K(f \cdot K(g \cdot h)) \neq K(f \cdot g \cdot h)$ — that *is* the obstruction.

9. The instance: the normalized-kernel magmoid **Norm**

The abstract home is inhabited by the very object that forced the question.

Proposition 9.1 (**Norm** is a normalized magmoid)**.** *Let $\mathcal{A}$ be the category of* substochastic kernels*: objects finite sets; morphisms nonnegative $X \times Y$ matrices with row sums ≤ 1 (sub-Markov kernels, $= X \to DMY$ for the subdistribution monad DM); composition = matrix product. Let K be* row-normalization*, $K(M)_{xy} = M_{xy}/r_x$ with $r_x = \sum_{y'} M_{xy'}$, acting as the identity on zero-mass (failure/$\perp$) rows. Then* $\operatorname{Im} K$ *is the set of normalized kernels $X \to MDY$ (each input a probability distribution of mass* 1*, or total failure $\perp$ of mass* 0*), and $N_K =$ **Norm** with $f \# g = K(f \cdot g)$ the Di Lavore–Román–Székely composition.*

Lemma 9.2 ((LA) holds for row-normalization)**.** *$K(K(x) \cdot c) = K(x \cdot c)$ for all composable substochastic x, c.*

Proof. On positive-mass rows $K(M) = D_M \cdot M$ with $D_M = \operatorname{diag}(1/r_i)$ a *positive diagonal*. Two observations. *(1) Positive-diagonal invariance:* for any positive diagonal $P = \operatorname{diag}(p_i)$, the matrix $(P \cdot M)_{ij} = p_i M_{ij}$ has row sum $p_i r_i$, so $K(P \cdot M)_{ij} = p_i M_{ij}/(p_i r_i) = M_{ij}/r_i = K(M)_{ij}$; hence $K(P \cdot M) = K(M)$. *(2)* By associativity of matrix product, $K(x) \cdot c = (D_x \cdot x) \cdot c = D_x \cdot (x \cdot c)$. Therefore $K(K(x) \cdot c) = K\big(D_x \cdot (x \cdot c)\big) \overset{(1)}{=} K(x \cdot c)$. On zero-mass rows $K(x) = x$ and both sides agree trivially. □

Note K is *not* left-multiplication by any *fixed* element: D_x depends on x. So (LA) is a genuine property here, not an identity forced by $k^2 = k$ — a point that becomes sharp in §10.

Lemma 9.3 (($\mathsf{RA}_\#$) fails; exact witness)**.** *There exist $f, g, h \in$ **Norm** with $K(f \cdot K(g \cdot h)) \neq K(f \cdot g \cdot h)$, so by Theorem 8.2* **Norm** *is non-associative.*

Proof. Over state sets of sizes $1 \to 2 \to 3 \to 2$, take the normalized kernels (every row of mass 1 or 0)

$$f = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \end{bmatrix}, \qquad g = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \qquad h = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{(middle state } 1 = \perp\text{)}.$$

Then $g \cdot h = \begin{bmatrix} \frac{2}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ has rows of *unequal* mass $\frac{2}{3}, \frac{1}{2}$ (the failure $\perp$ at the middle state destroys different amounts of mass on g's two rows). Computing,

$$(f \# g) \# h = K(f \cdot g \cdot h) = \begin{bmatrix} \frac{8}{11} & \frac{3}{11} \end{bmatrix} \quad \text{(Theorem 8.1)}, \qquad f \# (g \# h) = K\big(f \cdot K(g \cdot h)\big) = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \end{bmatrix}.$$

These differ, so $(\mathsf{RA}_{\#})$ fails. □

The mechanism is exactly Corollary 8.3 made concrete. The inner normalization $K(g \cdot h) = D_{g \cdot h} \cdot (g \cdot h)$ with $D_{g \cdot h} = \mathrm{diag}(\frac{3}{2}, 2)$ rescales the two middle rows by *different* factors $\frac{3}{2} \neq 2$; sandwiched between f and $g \cdot h$ it reweights the contraction and is not removed by the outer K. Had the two middle rows lost equal mass ($D_{g \cdot h} \propto I$) the defect would vanish — which is why a **Norm**-defect needs a middle contraction of dimension ≥ 3 (see §10).

Remark 9.4 (computational corroboration). A deterministic sweep (seed 20261019; 20000 trials) confirms the picture: (LA) violations 0/20000; Theorem 8.1 ($(f \# g) \# h = K(f \cdot g \cdot h)$) violations 0/20000; generic triples non-associative; and the two *recovery* regimes predicted by Theorem 8.2 — g, h stochastic so $D_{g \cdot h} = I$, and middle dimension 1 — associative 5000/5000. This reproduces arXiv:2510.01131's Proposition 4.9 purity cases exactly (each purity condition drives the defect to $\sim 2 \times 10^{-16}$, its negation to $\sim 10^{-2}$). These are *computed* corroborations; the proved content is Theorems 8.1–8.2 and Lemmas 9.2–9.3.

10. Minimality: the home is operator-level, and the fourth map falls

The dream that launched this search named the classical template as a *near-semiring with a one-sided absorbing zero* 0 ($0 \cdot a = 0$ imposed, $a \cdot 0 = 0$ not), with "the strict zero replaced by an idempotent". The natural first guess for the home is then an *element*: a distinguished idempotent κ with the normalizer $K(x) = \kappa \cdot x$. This is the fourth named map. We show it *provably cannot work* — and so building the home refutes it.

Call a normalizer *principal* if $K(x) = \kappa \cdot x$ for a fixed $\kappa \in \mathcal{A}$ with $\kappa \cdot \kappa = \kappa$, and *genuine* otherwise.

Proposition 10.1 (principal normalizers are associativity-preserving)**.** *If $K(x) = \kappa \cdot x$ with $\kappa \cdot \kappa = \kappa$, then (LA) holds automatically* and N_K *is* associative *— always. A distinguished idempotent* element *cannot carry handedness.*

Proof. (LA) is free: $K(K(x) \cdot c) = \kappa(\kappa x)c = (\kappa\kappa)xc = \kappa xc = K(x \cdot c)$. For associativity, note $\kappa \cdot g = g$ for $g \in \mathrm{Im}\, K$ (if $g = \kappa g'$ then $\kappa g = \kappa\kappa g' = \kappa g' = g$). Hence

$$f \# (g \# h) = \kappa \cdot f \cdot (\kappa \cdot g \cdot h) = (\kappa f) \cdot (\kappa g) \cdot h = (\kappa f) \cdot g \cdot h = \kappa \cdot f \cdot g \cdot h,$$

while $(f \# g) \# h = \kappa \cdot (\kappa \cdot f \cdot g) \cdot h = (\kappa\kappa) \cdot f \cdot g \cdot h = \kappa \cdot f \cdot g \cdot h$. The two agree. □

In Theorem 8.2's terms a principal normalizer *always* satisfies $(\mathsf{RA}_{\#})$: the inner κ in $f \cdot K(g \cdot h) = f \cdot \kappa \cdot (g \cdot h)$ meets g on its left, and $\kappa g = g$, so it is silently absorbed. The entire handedness phenomenon is *impossible* for $K = \kappa \cdot (-)$.

Corollary 10.2 (handedness requires a genuine, non-principal normalizer)**.** *If N_K is non-associative then K is* not *$\kappa \cdot (-)$ for any fixed κ. The row-normalization K of* **Norm** *is genuine: it is* non-linear *($K(\lambda M) = K(M) \neq \lambda K(M)$), whereas every principal K is linear.*

Proposition 10.1 is the fourth refutation, and it is the sharpest of the four because it is not about a failed map into some theory but about the *kind* of object handedness can live on.

Remark 10.3 (the triple-elimination, and why the near-semiring is only a shadow). The four named maps probe four loci, and each is now seen to be *inert* for handedness:

- *cochain / single morphism* (cocycle, §4) — the apparatus is defined by associativity;

- *single idempotent morphism* (Karoubi, §5) — k is two-sided tame as a morphism ($k\,;n = n$ *and* $m\,;k = m$), and splitting preserves associativity;
- *coherence 2-cell* (lax-DL, §6) — a fixed-direction $\leq$ cannot separate $\leq$-incomparable outputs;
- *distinguished element* (near-semiring zero, this section) — an idempotent *element* is always two-sidedly absorbing (Prop. 10.1), hence associativity-preserving.

Morphism, cell, and element loci are all inert; what remains is the *argument slots of the binary operation* #, and that is where Definition 7.1 places K. The classical near-semiring is a recognizable *shadow* — its one-sided distributivity has the right *shape* — but it carries its one-sidedness in an *element* (the zero) over extra data (the + that # does not need), and Proposition 10.1 proves an element can never be the source of non-associativity. Handedness is operator-level; its home is original ground.

Remark 10.4 (machine-checked, axiom-free). The construction is certified by a finite model (registry leaf `lean-positive-home-oracle`, **Lean-verified**). Since row-normalization uses rational division — not kernel-decidable — the **Norm** matrix instance stays computational (Python); to `decide` (LA) as a *universal* law and witness $\neg(\mathsf{RA}_{\#})$, we formalise the *smallest* abstract monoid-with-normalizer realizing $(\mathsf{LA})\wedge\neg(\mathsf{RA}_{\#})$, found by exhaustive search (none exists at order ≤ 3): $M = \{e, z, a, b\}$ with two-sided zero z, $a\cdot a = z$, $a\cdot b = a$, $b\cdot b = b$, and $K = \mathrm{id}$ on $\{e, a, b\}$, $K(z) = b$. Lean `decide` checks, over this model: M is a monoid; $(\mathsf{K1})$, (U); (LA) *universally*; Theorem 8.1; that $(\mathsf{RA}_{\#})$ *fails* and # is non-associative (witness a, a, a: $(a \,\#\, a) \,\#\, a = b \neq a = a \,\#\, (a \,\#\, a)$); Theorem 8.2 pointwise; the magmoid axioms (Remark 7.2); Corollary 10.2 (K non-principal); and the Proposition 10.1 control (every idempotent-element principal normalizer is associative). An `#print axioms` audit of the bundled oracle shows `[propext]` only — no `sorry`, no `Classical.choice`, no `native_decide`. This is the positive twin of the Karoubi-vacuity leaf `lean-split-idempotent-oracle`: together they make "genuine obstruction versus vacuous obstruction" machine-checked.

11. What the four maps missed, and why

The four refutations are not four coincidences; they are one structural fact read four ways. The cocycle and Karoubi homes are the *same fact read twice* — two independent obstruction theories, group-cohomological (H^2/cocycle) and Cauchy/Karoubi (idempotent-splitting), both *orthogonal* to associativity, and both say so because their apparatus is *defined by* or *preserves* associativity. The lax-DL and near-semiring homes extend this: a fixed-direction order cannot see a defect between incomparable outputs, and a distinguished element is always two-sidedly absorbing. In all four, the candidate lives at a locus — cochain, idempotent morphism, coherence cell, element — at which handedness is invisible. It lives one level up, at the argument slots of #, exactly where Theorem 8.2 places it. The mismatch is structural, not accidental: this is *why* the maps missed.

named map	locus it probes	verdict	kind of negative
$k \mapsto$ 2-cocycle	cochain/morphism	refuted	no receptacle
$k \mapsto$ Karoubi splitting	idempotent morphism	refuted	vacuous receptacle
$k \mapsto$ lax distributive law	coherence 2-cell	refuted	wrong locus ($\leq$-incomparable)
near-semiring zero	distinguished element	refuted	associativity-preserving
normalizer K	**argument slots of #**	**receives it**	**the home**

Remark 11.1 (originality: evidence, not theorem). A dedicated literature sweep for a prior categorical home of "one-sided absorptivity of a single distinguished map" found none. The two genuine precedents are algebra/ideal-flavored, structurally one level off: *near-semirings* (a one-sided absorbing zero — an element, refuted above by Proposition 10.1), and Grandis, "Categories with Negation" (arXiv:1709.03186), whose left/right-absorbing morphism *classes* form an *ideal*, not a single operator. We report this as the current state of evidence and would retract it the moment a genuine precedent surfaced.

12. Grant relevance

The program's obstruction-theory narrative gains a complete, honestly-bounded unit. For the central compositional tool — the Zappa–Szép weld of directed containers — we now know precisely *what the non-associativity is not*: not a cocycle, not a Karoubi-splitting obstruction, not a lax distributive law, and not a near-semiring zero; four named maps, decided, of four distinct kinds, with an independent confirmation that the Karoubi vacuity is generic. And we know *what it is*: a one-sided absorptivity (handedness) datum received by an idempotent normalizer K, with "associativity = K two-sided absorbing" (Theorem 8.2) a theorem rather than a slogan, with the normalized-kernel magmoid **Norm** as a faithful instance, with the home proved *operator-level* rather than element-level (Proposition 10.1), and with the abstract construction machine-checked axiom-free (Remark 10.4). "Handedness datum, not characteristic class" is a crisp, defensible statement about where compositional non-associativity lives — and the kind of negative-plus-positive result that keeps the Impact narrative ("obstruction = cohomology") honest about which obstructions are actually ours, and which home is new ground.

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